

Since  $\theta_0: H^n(X) \rightarrow H^n(X)$  is a cohomology operation that preserves dimension, it must be defined by a coefficient homomorphism  $\mathbb{Z}_p \rightarrow \mathbb{Z}_p$ , multiplication by some  $a_n \in \mathbb{Z}_p$ . We claim that these  $a_n$ 's satisfy

$$a_{m+n} = (-1)^{p(p-1)mn/2} a_m a_n \quad \text{and} \quad a_n = (-1)^{p(p-1)n(n-1)/4} a_1^n$$

To see this, recall the formula  $\lambda(\alpha \smile \beta) = (-1)^{p(p-1)mn/2} \lambda(\alpha) \lambda(\beta)$  for  $|\alpha| = m$  and  $|\beta| = n$ . From the definition of the  $\theta_i$ 's it then follows that  $\theta_0(\alpha \smile \beta) = (-1)^{p(p-1)mn/2} \theta_0(\alpha) \theta_0(\beta)$ , which gives the first part of the claim. The second part follows from this by induction on  $n$ .

|| **Lemma 4L.15.**  $a_1 = \pm m!$  for  $m = (p-1)/2$ , so  $p = 2m+1$ .

**Proof:** It suffices to compute  $\theta_0(\alpha)$  where  $\alpha$  is any nonzero 1-dimensional class, so the simplest thing is to choose  $\alpha$  to be a generator of  $H^1(S^1)$ , say a generator coming from a generator of  $H^1(S^1; \mathbb{Z})$ . This determines  $\alpha$  up to a sign. Since  $H^i(S^1) = 0$  for  $i > 1$ , we have  $\theta_i(\alpha) = 0$  for  $i > 0$ , so the defining formula for  $\theta_0(\alpha)$  has the form  $\nabla^*(\lambda(\alpha)) = \omega_{p-1} \otimes \theta_0(\alpha) = a_1 \omega_{p-1} \otimes \alpha$  in  $H^p(L^\infty \times S^1)$ . To compute  $a_1$  there is no harm in replacing  $L^\infty$  by a finite-dimensional lens space, say  $L^p$ , the  $p$ -skeleton of  $L^\infty$ . Thus we may restrict the bundle  $\Lambda S^1 \rightarrow L^\infty$  to a bundle  $\Lambda^p S^1 \rightarrow L^p$  with the same fibers  $(S^1)^{\wedge p} = S^p$ . We regard  $S^1$  as the one-point compactification of  $\mathbb{R}$  with basepoint the added point at infinity, and then  $(S^1)^{\wedge p}$  becomes the one-point compactification of  $\mathbb{R}^p$  with  $\mathbb{Z}_p$  acting by permuting the coordinates of  $\mathbb{R}^p$  cyclically, preserving the origin and the point at infinity. This action defines the bundle  $\Gamma^p S^1 \rightarrow L^p$  with fibers  $S^p$ , containing a zero section and a section at infinity, and  $\Lambda^p S^1$  is obtained by collapsing the section at infinity. We can also describe  $\Lambda^p S^1$  as the one-point compactification of the complement of the section at infinity in  $\Gamma^p S^1$ , since the base space  $L^p$  is compact. The complement of the section at infinity is a bundle  $E \rightarrow L^p$  with fibers  $\mathbb{R}^p$ . In general, the one-point compactification of a fiber bundle  $E$  over a compact base space with fibers  $\mathbb{R}^n$  is called the **Thom space**  $T(E)$  of the bundle, and a class in  $H^n(T(E))$  that restricts to a generator of  $H^n$  of the one-point compactification of each fiber  $\mathbb{R}^n$  is called a **Thom class**. In our situation,  $\lambda(\alpha)$  is such a Thom class.

Our first task is to construct subbundles  $E_0, E_1, \dots, E_m$  of  $E$ , where  $E_0$  has fiber  $\mathbb{R}$  and the other  $E_j$ 's have fiber  $\mathbb{R}^2$ , so  $p = 2m+1$ . The bundle  $E$  comes from the linear transformation  $T: \mathbb{R}^p \rightarrow \mathbb{R}^p$  permuting the coordinates cyclically. We claim there is a decomposition  $\mathbb{R}^p = V_0 \oplus V_1 \oplus \dots \oplus V_m$  with  $V_0$  1-dimensional and the other  $V_j$ 's 2-dimensional, such that  $T(V_j) = V_j$  for all  $j$ , with  $T|_{V_0}$  the identity and  $T|_{V_j}$  a rotation by the angle  $2\pi j/p$  for  $j > 0$ . Thus  $T$  defines an action of  $\mathbb{Z}_p$  on  $V_j$  and we can define  $E_j$  just as  $E$  was defined, as the quotient  $(S^p \times V_j)/\mathbb{Z}_p$  with respect to the diagonal action.

An easy way to get the decomposition  $\mathbb{R}^p = V_0 \oplus V_1 \oplus \dots \oplus V_m$  is to regard  $\mathbb{R}^p$  as a module over the principal ideal domain  $\mathbb{R}[t]$  by setting  $tv = T(v)$  for

$v \in \mathbb{R}^p$ . Then  $\mathbb{R}^p$  is isomorphic as a module to the module  $\mathbb{R}[t]/(t^p - 1)$  since  $T$  permutes coordinates cyclically; this amounts to identifying the standard basis vectors  $v_1, \dots, v_p$  in  $\mathbb{R}^p$  with  $1, t, \dots, t^{p-1}$ . The polynomial  $t^p - 1$  factors over  $\mathbb{C}$  into the linear factors  $t - e^{2\pi i j/p}$  for  $j = 0, \dots, p-1$ . Combining complex conjugate factors, this gives a factorization over  $\mathbb{R}$ ,  $t^p - 1 = (t-1) \prod_{1 \leq j \leq m} (t^2 - 2(\cos \varphi_j)t + 1)$ , where  $\varphi_j = 2\pi j/p$ . These are distinct monic irreducible factors, so the module  $\mathbb{R}[t]/(t^p - 1)$  splits as  $\mathbb{R}[t]/(t-1) \oplus_{1 \leq j \leq m} \mathbb{R}[t]/(t^2 - 2(\cos \varphi_j)t + 1)$  by the basic structure theory of modules over a principal ideal domain. This translates into a decomposition  $\mathbb{R}^p = V_0 \oplus V_1 \oplus \dots \oplus V_m$  with  $T(V_j) \subset V_j$ . Here  $V_0$  corresponds to  $\mathbb{R}[t]/(t-1) \approx \mathbb{R}$  with  $t$  acting as the identity, and  $V_j$  for  $j > 0$  corresponds to  $\mathbb{R}[t]/(t^2 - 2(\cos \varphi_j)t + 1)$ . The latter module is isomorphic to  $\mathbb{R}^2$  with  $t$  acting as rotation by the angle  $\varphi_j$  since the characteristic polynomial of this rotation is readily computed to be  $t^2 - 2(\cos \varphi_j)t + 1$ , hence this rotation satisfies  $t^2 - 2(\cos \varphi_j)t + 1 = 0$  so there is a module homomorphism  $\mathbb{R}[t]/(t^2 - 2(\cos \varphi_j)t + 1) \rightarrow \mathbb{R}^2$  which is obviously an isomorphism.

From the decomposition  $\mathbb{R}^p = V_0 \oplus V_1 \oplus \dots \oplus V_m$  and the action of  $T$  on each factor we can see that the only vectors fixed by  $T$  are those in the line  $V_0$ . The vectors  $(x, \dots, x)$  are fixed by  $T$ , so  $V_0$  must be this diagonal line.

Next we compute Thom classes for the bundles  $E_j$ . This is easy for  $E_0$  which is the product  $L^p \times \mathbb{R}$ , so the projection  $E_0 \rightarrow \mathbb{R}$  one-point compactifies to a map  $T(E_0) \rightarrow S^1$  and we can pull back the chosen generator  $\alpha \in H^1(S^1)$  to a Thom class for  $E_0$ . The other  $E_j$ 's have 2-dimensional fibers, which we now view as  $\mathbb{C}$  rather than  $\mathbb{R}^2$ . Just as  $E_j$  is the quotient of  $S^p \times \mathbb{C}$  via the identifications  $(v, z) \sim (e^{2\pi i/p} v, e^{2\pi i j/p} z)$ , we can define a bundle  $\bar{E}_j \rightarrow \mathbb{C}P^m$  with fiber  $\mathbb{C}$  by the identifications  $(v, z) \sim (\lambda v, \lambda^j z)$  for  $\lambda \in S^1 \subset \mathbb{C}$ . We then have the left half of the commutative diagram shown at the right, where the quotient map  $\tilde{q}$  restricts to a homeomorphism on each fiber. The maps  $\tilde{f}$  and  $f$  are induced by the map  $S^p \times \mathbb{C} \rightarrow S^p \times \mathbb{C}$  sending  $(v, z)$  to  $(v^j, z)$  where  $v^j$  means raise each coordinate of  $v$  to the  $j^{\text{th}}$  power and then rescale to get a vector of unit length. The map  $\tilde{f}$  is well-defined since equivalent pairs  $(v, z) \sim (\lambda v, \lambda^j z)$  in  $\bar{E}_j$  are carried to pairs  $(v^j, z)$  and  $(\lambda^j v^j, \lambda^j z)$  that are equivalent in  $\bar{E}_1$ .

$$\begin{array}{ccccc} E_j & \xrightarrow{\tilde{q}} & \bar{E}_j & \xrightarrow{\tilde{f}} & \bar{E}_1 \\ \downarrow & & \downarrow & & \downarrow \\ L^p & \xrightarrow{q} & \mathbb{C}P^m & \xrightarrow{f} & \mathbb{C}P^m \end{array}$$

Since both  $\tilde{q}$  and  $\tilde{f}$  restrict to homeomorphisms in each fiber, they extend to maps of Thom spaces that pull a Thom class for  $\bar{E}_1$  back to Thom classes for  $\bar{E}_j$  and  $E_j$ . To construct a Thom class for  $\bar{E}_1$ , observe that the Thom space  $T(\bar{E}_1)$  is homeomorphic to  $\mathbb{C}P^{m+1}$ , namely, view the sphere  $S^p = S^{2m+1}$  as the unit sphere in  $\mathbb{C}^{m+1}$ , and then the inclusion  $S^p \times \mathbb{C} \hookrightarrow \mathbb{C}^{m+1} \times \mathbb{C} = \mathbb{C}^{m+2}$  induces a map  $g: \bar{E}_1 \rightarrow \mathbb{C}P^{m+1}$  since the equivalence relation defining  $\bar{E}_1$  is  $(v, z) \sim (\lambda v, \lambda z)$  for  $\lambda \in S^1$ . It is evident that  $g$  is a homeomorphism onto the complement of the point  $[0, \dots, 0, 1]$  in  $\mathbb{C}P^{m+1}$ , so sending the point at infinity in  $T(\bar{E}_1)$  to  $[0, \dots, 0, 1]$  gives an extension

of  $g$  to a homeomorphism  $T(\bar{E}_1) \approx \mathbb{CP}^{m+1}$ . Under this homeomorphism the one-point compactifications of the fibers of  $\bar{E}_1$  correspond to the 2-spheres  $S_v^2$  consisting of  $[0, \dots, 0, 1]$  and the points  $[v, z] \in \mathbb{CP}^{m+1}$  with fixed  $v \in S^p$  and varying  $z \in \mathbb{C}$ . Each  $S_v^2$  is a  $\mathbb{CP}^1$  in  $\mathbb{CP}^{m+1}$  equivalent to the standard  $\mathbb{CP}^1$  under a homeomorphism of  $\mathbb{CP}^{m+1}$  coming from a linear isomorphism of  $\mathbb{C}^{m+2}$ , so a generator  $\gamma$  of  $H^2(\mathbb{CP}^{m+1})$  is a Thom class, restricting to a generator of  $H^2(S_v^2)$  for each  $v$ . We choose  $\gamma$  to be the  $\mathbb{Z}_p$  reduction of a generator of  $H^2(\mathbb{CP}^{m+1}; \mathbb{Z})$ , so  $\gamma$  is determined up to a sign.

A slightly different view of Thom classes will be useful. For the bundle  $E \rightarrow L^p$ , for example, we have isomorphisms

$$\begin{aligned} \tilde{H}^*(T(E)) &\approx H^*(T(E), \infty) \quad \text{where } \infty \text{ is the compactification point} \\ &\approx H^*(T(E), T(E) - L^p) \quad \text{where } L^p \text{ is embedded in } T(E) \text{ as the zero} \\ &\quad \text{section, so } T(E) - L^p \text{ deformation retracts onto } \infty \\ &\approx H^*(E, E - L^p) \quad \text{by excision} \end{aligned}$$

Thus we can view a Thom class as lying in  $H^*(E, E - L^p)$ , and similarly for the bundles  $E_j$ .

We have projections  $\pi_j: E \rightarrow E_j$  via the projections  $V_0 \oplus V_1 \oplus \dots \oplus V_m \rightarrow V_j$  in fibers. If  $\tau_j \in H^*(E_j, E_j - L^p)$  denotes the Thom class constructed above, then we have the pullback  $\pi_j^*(\tau_j) \in H^*(E, E - \pi_j^{-1}(L^p))$ , and the cup product  $\prod_j \pi_j^*(\tau_j)$  in  $H^*(E, E - L^p)$  is a Thom class for  $E$ , as one sees by applying the calculation at the end of Example 3.11 in each fiber. Under the isomorphism  $H^*(E, E - L^p) \approx \tilde{H}^*(T(E))$ , the class  $\prod_j \pi_j^*(\tau_j)$  corresponds to  $\pm \lambda(\alpha)$  since both classes restrict to  $\pm \alpha^{\otimes p}$  in each fiber  $S^p \subset T(E)$  and  $\lambda(\alpha)$  is uniquely determined by its restriction to fibers.

Now we can finish the proof of the lemma. The class  $\nabla^*(\lambda(\alpha))$  is obtained by restricting  $\lambda(\alpha) \in H^p(T(E))$  to the diagonal  $T(E_0)$ , then pulling back to  $L^p \times S^1$  via the quotient map  $L^p \times S^1 \rightarrow T(E_0)$  which collapses the section at infinity to a point. Restricting  $\prod_j \pi_j^*(\tau_j)$  to  $H^p(E_0, E_0 - L^p) \approx H^p(T(E_0))$  gives  $\tau_0 \smile e_1 \smile \dots \smile e_m$  where  $e_j \in H^2(E_0)$  is the image of  $\tau_j$  under  $H^2(E_j, E_j - L^p) \rightarrow H^2(E_j) \approx H^2(L^p) \approx H^2(E_0)$ , these last two isomorphisms coming from including  $L^p$  in  $E_j$  and  $E_0$  as the zero section, to which they deformation retract. To compute  $e_j$ , we use the diagram

$$\begin{array}{ccccc} H^2(E_j, E_j - L^p) & \xleftarrow{\tilde{q}^*} & H^2(\bar{E}_j, \bar{E}_j - \mathbb{CP}^m) & \xleftarrow{\tilde{f}^*} & H^2(\bar{E}_1, \bar{E}_1 - \mathbb{CP}^m) \\ \downarrow & & \downarrow & & \downarrow \\ H^2(E_j) & \xleftarrow{\tilde{q}^*} & H^2(\bar{E}_j) & \xleftarrow{\tilde{f}^*} & H^2(\bar{E}_1) \\ \downarrow \approx & & \downarrow \approx & & \downarrow \approx \\ H^2(L^p) & \xleftarrow{q^*} & H^2(\mathbb{CP}^m) & \xleftarrow{f^*} & H^2(\mathbb{CP}^m) \end{array}$$

The Thom class for  $\bar{E}_1$  lies in the upper right group. Following this class across the top of the diagram and then down to the lower left corner gives the element  $e_j$ . To compute  $e_j$  we take the alternate route through the lower right corner of the diagram. The image of the Thom class for  $\bar{E}_1$  in the lower right  $H^2(\mathbb{CP}^m)$  is the generator  $\gamma$

since  $T(\bar{E}_1) = \mathbb{CP}^{m+1}$ . The map  $f^*$  is multiplication by  $j$  since  $f$  has degree  $j$  on  $\mathbb{CP}^1 \subset \mathbb{CP}^m$ . And  $q^*(\gamma) = \pm\omega_2$  since  $q$  restricts to a homeomorphism on the 2-cell of  $L^p$  in the CW structure defined in Example 2.43. Thus  $e_j = \pm j\omega_2$ , and so  $\tau_0 \smile e_1 \smile \cdots \smile e_m = \pm m!\tau_0 \smile \omega_2^m = \pm m!\tau_0 \smile \omega_{p-1}$ . Since  $\tau_0$  was the pullback of  $\alpha$  via the projection  $T(E_0) \rightarrow S^1$ , when we pull  $\tau_0$  back to  $L^p \times S^1$  via  $\nabla$  we get  $1 \otimes \alpha$ , so  $\tau_0 \smile e_1 \smile \cdots \smile e_m$  pulls back to  $\pm m!\omega_{p-1} \otimes \alpha$ . Hence  $a_1 = \pm m!$ .  $\square$

The lemma implies in particular that  $a_n$  is not zero in  $\mathbb{Z}_p$ , so  $a_n$  has a multiplicative inverse  $a_n^{-1}$  in  $\mathbb{Z}_p$ . We then define

$$P^i(\alpha) = (-1)^i a_n^{-1} \theta_{2i(p-1)}(\alpha) \quad \text{for } \alpha \in H^n(X)$$

The factor  $a_n^{-1}$  guarantees that  $P^0$  is the identity. The factor  $(-1)^i$  is inserted in order to make  $P^i(\alpha) = \alpha^p$  if  $|\alpha| = 2i$ , as we show next. We know that  $\theta_{2i(p-1)}(\alpha) = \alpha^p$ , so what must be shown is that  $(-1)^i a_{2i}^{-1} = 1$ , or equivalently,  $a_{2i} = (-1)^i$ .

To do this we need a number theory fact:  $((p-1)/2)!^2 \equiv (-1)^{(p+1)/2} \pmod{p}$ . To derive this, note first that the product of all the elements  $\pm 1, \pm 2, \dots, \pm(p-1)/2$  of  $\mathbb{Z}_p^*$  is  $((p-1)/2)!^2 (-1)^{(p-1)/2}$ . On the other hand, this group is cyclic of even order, so the product of all its elements is the unique element of order 2, which is  $-1$ , since all the other nontrivial elements cancel their inverses in this product. Thus  $((p-1)/2)!^2 (-1)^{(p-1)/2} \equiv -1$  and hence  $((p-1)/2)!^2 \equiv (-1)^{(p+1)/2} \pmod{p}$ .

Using the formulas  $a_n = (-1)^{p(p-1)n(n-1)/4} a_1^n$  and  $a_1 = \pm((p-1)/2)!$  we then have

$$\begin{aligned} a_{2i} &= (-1)^{p(p-1)2i(2i-1)/4} ((p-1)/2)!^{2i} \\ &= (-1)^{p[(p-1)/2]i(2i-1)} (-1)^{i(p+1)/2} \\ &= (-1)^{i(p-1)/2} (-1)^{i(p+1)/2} \quad \text{since } p \text{ and } 2i-1 \text{ are odd} \\ &= (-1)^{ip} = (-1)^i \quad \text{since } p \text{ is odd.} \end{aligned}$$

**Theorem 4L.16.** *The operations  $P^i$  satisfy the properties (1)–(6) and the Adem relations.*

**Proof:** The first two properties, naturality and additivity, along with the fact that  $P^i(\alpha) = 0$  if  $2i > |\alpha|$ , are inherited from the  $\theta_i$ 's. Property (6) and the other half of (5) have just been shown above. For the Cartan formula we have, for  $\alpha \in H^m$  and  $\beta \in H^n$ ,  $\lambda(\alpha \smile \beta) = (-1)^{p(p-1)mn/2} \lambda(\alpha) \lambda(\beta)$  and hence

$$\begin{aligned} \sum_i \omega_{(p-1)(m+n)-i} \otimes \theta_i(\alpha \smile \beta) &= \\ (-1)^{p(p-1)mn/2} \left( \sum_j \omega_{(p-1)m-j} \otimes \theta_j(\alpha) \right) \left( \sum_k \omega_{(p-1)n-k} \otimes \theta_k(\beta) \right) \end{aligned}$$

Recall that  $\omega_{2r} = \omega_2^r$  and  $\omega_{2r+1} = \omega_1 \omega_2^r$ , with  $\omega_1^2 = 0$ . Therefore terms with  $i$  even on the left side of the equation can only come from terms with  $j$  and  $k$  even on the

right side. This leads to the second equality in the following sequence:

$$\begin{aligned}
 P^i(\alpha \smile \beta) &= (-1)^i a_{m+n}^{-1} \theta_{2i(p-1)}(\alpha \smile \beta) \\
 &= (-1)^i a_{m+n}^{-1} (-1)^{p(p-1)mn/2} \sum_j \theta_{2(i-j)(p-1)}(\alpha) \theta_{2j(p-1)}(\beta) \\
 &= \sum_j (-1)^{i-j} a_m^{-1} \theta_{2(i-j)(p-1)}(\alpha) (-1)^j a_n^{-1} \theta_{2j(p-1)}(\beta) \\
 &= \sum_j P^{i-j}(\alpha) P^j(\beta)
 \end{aligned}$$

Property (4), the invariance of  $P^i$  under suspension, follows from the Cartan formula just as in the case  $p = 2$ , using the fact that  $P^0$  is the only  $P^i$  that can be nonzero on 1-dimensional classes, by (5).

It remains to prove the Adem relations for Steenrod powers. We will need a Bockstein calculation:

**Lemma 4L.17.**  $\beta \theta_{2k} = -\theta_{2k+1}$ .

**Proof:** Let us first reduce the problem to showing that  $\beta \nabla^*(\lambda(\iota)) = 0$ . If we compute  $\beta \nabla^*(\lambda(\iota))$  using the product formula for  $\beta$ , we get

$$\beta \left( \sum_i \omega_{(p-1)n-i} \otimes \theta_i(\iota) \right) = \sum_i \left( \beta \omega_{(p-1)n-i} \otimes \theta_i(\iota) + (-1)^i \omega_{(p-1)n-i} \otimes \beta \theta_i(\iota) \right)$$

Since  $\beta \omega_{2j-1} = \omega_{2j}$  and  $\beta \omega_{2j} = 0$ , the terms with  $i = 2k$  and  $i = 2k+1$  give  $\sum_k \omega_{(p-1)n-2k} \otimes \beta \theta_{2k}(\iota)$  and  $\sum_k \omega_{(p-1)n-2k} \otimes \theta_{2k+1}(\iota) - \sum_k \omega_{(p-1)n-2k-1} \otimes \beta \theta_{2k+1}(\iota)$ , respectively. Thus the coefficient of  $\omega_{(p-1)n-2k}$  in  $\beta \nabla^*(\lambda(\iota))$  is  $\beta \theta_{2k}(\iota) + \theta_{2k+1}(\iota)$ , so if we assume that  $\beta \nabla^*(\lambda(\iota)) = 0$ , this coefficient must vanish since we are in the tensor product  $H^*(L^\infty) \otimes H^*(K_n)$ . So we get  $\beta \theta_{2k}(\iota) = -\theta_{2k+1}(\iota)$  and hence  $\beta \theta_{2k}(\alpha) = -\theta_{2k+1}(\alpha)$  for all  $\alpha$ . Note that  $\beta \theta_{2k+1} = 0$  from the coefficient of  $\omega_{(p-1)n-2k-1}$ . This also follows from the formula  $\beta \theta_{2k} = -\theta_{2k+1}$  since  $\beta^2 = 0$ .

In order to show that  $\beta \nabla^*(\lambda(\iota)) = 0$  we first compute  $\beta \lambda(\iota)$ . We may assume  $K_n$  has a single  $n$ -cell and a single  $(n+1)$ -cell, attached by a map of degree  $p$ . Let  $\varphi$  and  $\psi$  be the cellular cochains assigning the value 1 to the  $n$ -cell and the  $(n+1)$ -cell, respectively, so  $\delta \varphi = p\psi$ . In  $K_n^{\wedge p}$  we then have

$$(*) \quad \delta(\varphi^{\otimes p}) = \sum_i (-1)^{in} \varphi^{\otimes i} \otimes \delta \varphi \otimes \varphi^{\otimes (p-i-1)} = p \sum_i (-1)^{in} \varphi^{\otimes i} \otimes \psi \otimes \varphi^{\otimes (p-i-1)}$$

where the tensor notation means cellular cross product, so for example  $\varphi^{\otimes p}$  is the cellular cochain dual to the  $np$ -cell  $e^n \times \cdots \times e^n$  of  $K_n^{\wedge p}$ . The formula  $(*)$  holds also in  $\Delta K_n$  since the latter space has only one  $(np+1)$ -cell not in  $K_n^{\wedge p}$ , with cellular boundary zero. Namely, this cell is the product of the 1-cell of  $L^\infty$  and the  $np$ -cell of  $K_n^{\wedge p}$  with one end of this product attached to the  $np$ -cell by the identity map and the other end by the cyclic permutation  $T$ , which has degree  $+1$  since  $p$  is odd, so these two terms in the boundary of this cell cancel, and there are no other terms since the rest of the attachment of this cell is at the basepoint.

Bockstein homomorphisms can be computed using cellular cochain complexes, so the formula  $(*)$  says that  $\sum_i (-1)^{in} \varphi^{\otimes i} \otimes \psi \otimes \varphi^{\otimes (p-i-1)}$  represents  $\beta \lambda(\iota)$ . Via the

quotient map  $\Gamma K_n \rightarrow \Lambda K_n$ , the class  $\lambda(\iota)$  pulls back to a class  $\gamma(\iota)$  with  $\beta\gamma(\iota)$  also represented by  $\sum_i (-1)^{in} \varphi^{\otimes i} \otimes \psi \otimes \varphi^{\otimes(p-i-1)}$ . To see what happens when we pull  $\beta\gamma(\iota)$  back to  $\beta\nabla^*(\lambda(\iota))$  via the inclusion  $L^\infty \times K_n \hookrightarrow \Gamma K_n$ , consider the commutative diagram at the right. In the left-hand square the maps  $\pi^*$  are induced by the covering space projections  $\pi: S^\infty \times K_n^{\wedge p} \rightarrow \Gamma K_n$  and  $\pi: S^\infty \times K_n \rightarrow L^\infty \times K_n$  arising from the free  $\mathbb{Z}_p$  actions. The vertical maps are induced by the diagonal inclusion  $S^\infty \times K \hookrightarrow S^\infty \times K_n^{\wedge p}$ . The maps  $\tau$  are the transfer homomorphisms defined in §3.G. Recall the definition: If  $\pi: \tilde{X} \rightarrow X$  is a  $p$ -sheeted covering space, a chain map  $C_*(X) \rightarrow C_*(\tilde{X})$  is defined by sending a singular simplex  $\sigma: \Delta^k \rightarrow X$  to the sum of its  $p$  lifts to  $\tilde{X}$ , and  $\tau$  is the induced map on cohomology. The key property of  $\tau$  is that  $\tau\pi^*: H^*(X) \rightarrow H^*(\tilde{X})$  is multiplication by  $p$ , for any choice of coefficient group, since when we project the  $p$  lifts of  $\sigma: \Delta^k \rightarrow X$  back to  $X$  we get  $p\sigma$ . When  $X$  is a CW complex and  $\tilde{X}$  is given the lifted CW structure, then  $\tau$  can also be defined in cellular cohomology by the same procedure.

Let us compute the value of the upper  $\tau$  in the diagram on  $1 \otimes \psi \otimes \varphi^{\otimes(p-1)}$  where ‘1’ is the cellular cocycle assigning the value 1 to each 0-cell of  $S^\infty$ . By the definition of  $\tau$  we have  $\tau(1 \otimes \psi \otimes \varphi^{\otimes(p-1)}) = \sum_i T^i(\psi \otimes \varphi^{\otimes(p-1)})$  where  $T: K_n^{\wedge p} \rightarrow K_n^{\wedge p}$  permutes the factors cyclically. It does not matter whether  $T$  moves coordinates one unit leftwards or one unit rightwards since we are summing over all the powers of  $T$ , so let us say  $T$  moves coordinates rightward. Then  $T(\psi \otimes \varphi^{\otimes(p-1)}) = \varphi \otimes \psi \otimes \varphi^{\otimes(p-2)}$ , with the last  $\varphi$  moved into the first position. This move is achieved by transposing this  $\varphi$  with each of the preceding  $p-2$   $\varphi$ ’s and with  $\psi$ . Transposing two  $\varphi$ ’s introduces a sign  $(-1)^{n^2}$ , and transposing  $\varphi$  with  $\psi$  introduces a sign  $(-1)^{n(n+1)} = +1$ , by the commutativity property of cross product. Thus the total sign introduced by  $T$  is  $(-1)^{n^2(p-2)}$ , which equals  $(-1)^n$  since  $p$  is odd. Each successive iterate of  $T$  also introduces a sign of  $(-1)^n$ , so  $T^i$  introduces a sign  $(-1)^{in}$  for  $0 \leq i \leq p-1$ . Thus

$$\tau(1 \otimes \psi \otimes \varphi^{\otimes(p-1)}) = \sum_i T^i(\psi \otimes \varphi^{\otimes(p-1)}) = \sum_i (-1)^{in} \varphi^{\otimes i} \otimes \psi \otimes \varphi^{\otimes(p-i-1)}$$

As observed earlier, this last cocycle represents the class  $\beta\gamma(\iota)$ .

Since  $\beta\gamma(\iota)$  is in the image of the upper  $\tau$  in the diagram, the image of  $\beta\gamma(\iota)$  in  $H^*(L^\infty \times K_n)$ , which is  $\nabla^*(\beta\lambda(\iota))$ , is in the image of the lower  $\tau$  since the right-hand square commutes. The map  $\pi^*$  in the lower row is obviously onto since  $S^\infty$  is contractible, so  $\nabla^*(\beta\lambda(\iota))$  is in the image of the composition  $\tau\pi^*$  across the bottom of the diagram. But this composition is multiplication by  $p$ , which is zero for  $\mathbb{Z}_p$  coefficients, so  $\beta\nabla^*(\lambda(\iota)) = \nabla^*(\beta\lambda(\iota)) = 0$ .  $\square$

The derivation of the Adem relations now follows the pattern for the case  $p = 2$ . We had the formula  $\nabla_2^* \lambda_2^*(\iota) = \sum_{i,j} \omega_{(p-1)pn-i} \otimes \theta_i(\omega_{(p-1)n-j} \otimes \theta_j(\iota))$ . Since we are

letting  $p = 2m + 1$ , this can be rewritten as  $\sum_{i,j} \omega_{2mpn-i} \otimes \theta_i(\omega_{2mn-j} \otimes \theta_j(\iota))$ . The only nonzero  $\theta_i$ 's are  $\theta_{2i(p-1)} = (-1)^i a_n P^i$  and  $\theta_{2i(p-1)+1} = -\beta \theta_{2i(p-1)}$  so we have

$$\begin{aligned} \sum_{i,j} \omega_{2mpn-i} \otimes \theta_i(\omega_{2mn-j} \otimes \theta_j(\iota)) = & \\ & \sum_{i,j} (-1)^{i+j} a_{2mn} a_n \omega_{2m(pn-2i)} \otimes P^i(\omega_{2m(n-2j)} \otimes P^j(\iota)) \\ & - \sum_{i,j} (-1)^{i+j} a_{2mn} a_n \omega_{2m(pn-2i)} \otimes P^i(\omega_{2m(n-2j)-1} \otimes \beta P^j(\iota)) \\ & - \sum_{i,j} (-1)^{i+j} a_{2mn} a_n \omega_{2m(pn-2i)-1} \otimes \beta P^i(\omega_{2m(n-2j)} \otimes P^j(\iota)) \\ & + \sum_{i,j} (-1)^{i+j} a_{2mn} a_n \omega_{2m(pn-2i)-1} \otimes \beta P^i(\omega_{2m(n-2j)-1} \otimes \beta P^j(\iota)) \end{aligned}$$

Since  $m$  and  $n$  will be fixed throughout the discussion, we may factor out the nonzero constant  $a_{2mn} a_n$ . Then applying the Cartan formula to expand the  $P^i$  terms, using also the formulas  $P^k(\omega_{2r}) = \binom{r}{k} \omega_{2r+2k(p-1)}$  and  $P^k(\omega_{2r+1}) = \binom{r}{k} \omega_{2r+2k(p-1)+1}$  derived earlier in the section, we obtain

$$\begin{aligned} & \sum_{i,j,k} (-1)^{i+j} \binom{m(n-2j)}{k} \omega_{2m(pn-2i)} \otimes \omega_{2m(n-2j+2k)} \otimes P^{i-k} P^j(\iota) \\ & - \sum_{i,j,k} (-1)^{i+j} \binom{m(n-2j)-1}{k} \omega_{2m(pn-2i)} \otimes \omega_{2m(n-2j+2k)-1} \otimes P^{i-k} \beta P^j(\iota) \\ & - \sum_{i,j,k} (-1)^{i+j} \binom{m(n-2j)}{k} \omega_{2m(pn-2i)-1} \otimes \omega_{2m(n-2j+2k)} \otimes \beta P^{i-k} P^j(\iota) \\ & + \sum_{i,j,k} (-1)^{i+j} \binom{m(n-2j)-1}{k} \omega_{2m(pn-2i)-1} \otimes \omega_{2m(n-2j+2k)} \otimes P^{i-k} \beta P^j(\iota) \\ & - \sum_{i,j,k} (-1)^{i+j} \binom{m(n-2j)-1}{k} \omega_{2m(pn-2i)-1} \otimes \omega_{2m(n-2j+2k)-1} \otimes \beta P^{i-k} \beta P^j(\iota) \end{aligned}$$

Letting  $\ell = mn + j - k$ , so that  $n - 2j + 2k = pn - 2\ell$ , the first of these five summations becomes

$$\sum_{i,j,\ell} (-1)^{i+j} \binom{m(n-2j)}{mn+j-\ell} \omega_{2m(pn-2i)} \otimes \omega_{2m(pn-2\ell)} \otimes P^{i+\ell-mn-j} P^j(\iota)$$

and similarly for the other four summations.

Now we bring in the symmetry property  $\varphi_{rs} = (-1)^{rs+mn} \varphi_{sr}$ , where, as before,  $\nabla_2^* \lambda_2^*(\iota) = \sum_{r,s} \omega_r \otimes \omega_s \otimes \varphi_{rs}$ . Of the five summations, only the first has both  $\omega$  terms with even subscripts, namely  $r = 2m(pn - 2i)$  and  $s = 2m(pn - 2\ell)$ , so the coefficient of  $\omega_r \otimes \omega_s$  in this summation must be symmetric with respect to switching  $i$  and  $\ell$ , up to a sign which will be  $+$  if we choose  $n$  to be even, as we will do. This gives the relation

$$(1) \quad \sum_j (-1)^{i+j} \binom{m(n-2j)}{mn+j-\ell} P^{i+\ell-mn-j} P^j(\iota) = \sum_j (-1)^{\ell+j} \binom{m(n-2j)}{mn+j-i} P^{i+\ell-mn-j} P^j(\iota)$$

Similarly, the second, third, and fourth summations involve  $\omega$ 's with subscripts of opposite parity, yielding the relation

$$(2) \quad \begin{aligned} & \sum_j (-1)^{i+j} \binom{m(n-2j)-1}{mn+j-\ell} P^{i+\ell-mn-j} \beta P^j(\iota) = \\ & \sum_j (-1)^{\ell+j} \binom{m(n-2j)-1}{mn+j-i} \beta P^{i+\ell-mn-j} P^j(\iota) - \sum_j (-1)^{\ell+j} \binom{m(n-2j)-1}{mn+j-i} P^{i+\ell-mn-j} \beta P^j(\iota) \end{aligned}$$

The relations (1) and (2) will yield the two Adem relations, so we will not need to consider the relation arising from the fifth summation.

To get the first Adem relation from (1) we choose  $n$  and  $\ell$  so that the left side of (1) has only one term, namely we take  $n = 2(1 + p + \cdots + p^{r-1}) + 2s$  and  $\ell = mn + s$  for given integers  $r$  and  $s$ . Then

$$\binom{m(n-2j)}{mn+j-\ell} = \binom{p^{r-1}-(p-1)(j-s)}{j-s}$$

and if  $r$  is large, this binomial coefficient is 1 if  $j = s$  and 0 otherwise since if the rightmost nonzero digit in the  $p$ -adic expansion of the 'denominator'  $j - s$  is  $x$ , then the corresponding digit of the 'numerator'  $(p-1)[(1+p+\cdots+p^{r-1}) - (j-s)]$  is obtained by reducing  $(p-1)(1-x) \bmod p$ , giving  $x-1$ , and  $\binom{x-1}{x} = 0$ . Then (1) becomes

$$\begin{aligned} (-1)^{i+s} P^i P^s(\iota) &= \sum_j (-1)^{\ell+j} \binom{m(n-2j)}{mn+j-i} P^{i+s-j} P^j(\iota) \\ \text{or } P^i P^s(\iota) &= \sum_j (-1)^{i+j} \binom{m(n-2j)}{mn+j-i} P^{i+s-j} P^j(\iota) \quad \text{since } \ell \equiv s \bmod 2 \\ &= \sum_j (-1)^{i+j} \binom{m(n-2j)}{i-pj} P^{i+s-j} P^j(\iota) \quad \text{since } \binom{x}{y} = \binom{x}{x-y} \\ &= \sum_j (-1)^{i+j} \binom{p^{r+(p-1)(s-j)-1}}{i-pj} P^{i+s-j} P^j(\iota) \end{aligned}$$

If  $r$  is large and  $i < ps$ , the term  $p^r$  in the binomial coefficient can be omitted since we may assume  $i \geq pj$ , hence  $j < s$ , so  $-1 + (p-1)(s-j) \geq 0$  and the  $p^r$  has no effect on the binomial coefficient if  $r$  is large. This shows the first Adem relation holds for the class  $\iota$ , and the general case follows as in the case  $p = 2$ .

To get the second Adem relation we choose  $n = 2p^r + 2s$  and  $\ell = mn + s$ . Reasoning as before, the left side of (2) then reduces to  $(-1)^{i+s} P^i \beta P^s(\iota)$  and (2) becomes

$$\begin{aligned} P^i \beta P^s(\iota) &= \sum_j (-1)^{i+j} \binom{(p-1)(p^r+s-j)}{i-pj} \beta P^{i+s-j} P^j(\iota) \\ &\quad - \sum_j (-1)^{i+j} \binom{(p-1)(p^r+s-j)-1}{i-pj-1} P^{i+s-j} \beta P^j(\iota) \end{aligned}$$

This time the term  $p^r$  can be omitted if  $r$  is large and  $i \leq ps$ . □

## Exercises

1. Determine all cohomology operations  $H^1(X; \mathbb{Z}) \rightarrow H^n(X; \mathbb{Z})$ ,  $H^2(X; \mathbb{Z}) \rightarrow H^n(X; \mathbb{Z})$ , and  $H^1(X; \mathbb{Z}_p) \rightarrow H^n(X; \mathbb{Z}_p)$  for  $p$  prime.
2. Use cohomology operations to show that the spaces  $(S^1 \times \mathbb{CP}^\infty)/(S^1 \times \{x_0\})$  and  $S^3 \times \mathbb{CP}^\infty$  are not homotopy equivalent.
3. Since there is a fiber bundle  $S^2 \rightarrow \mathbb{CP}^5 \rightarrow \mathbb{HP}^2$  by Exercise 35 in §4.2, one might ask whether there is an analogous bundle  $S^4 \rightarrow \mathbb{HP}^5 \rightarrow \mathbb{OP}^2$ . Use Steenrod powers for the prime 3 to show that such a bundle cannot exist. [The Gysin sequence can be used to determine the map on cohomology induced by the bundle projection  $\mathbb{HP}^5 \rightarrow \mathbb{OP}^2$ .]
4. Show there is no fiber bundle  $S^7 \rightarrow S^{23} \rightarrow \mathbb{OP}^2$ . [Compute the cohomology ring of the mapping cone of the projection  $S^{23} \rightarrow \mathbb{OP}^2$  via Poincaré duality or the Thom isomorphism.]